

The 67th ASH Annual Meeting Abstracts

POSTER

701. EXPERIMENTAL TRANSPLANTATION: BASIC AND TRANSLATIONAL

Multiplex detection of gastrointestinal microbiota using short probes in graft vs. host disease

Albert Yeh¹, Annalyssa Long¹, David Sierra¹, Paul Kong¹, Kathleen Ensbey¹, Samuel Legg¹, Tomoko Sekiguchi¹, Tomas Bencomo¹, Anna Elz¹, Neelendu Dey¹, Armin Rashidi¹, Kate Markey¹, David Fredricks¹, David Hockenbery¹, Geoff Hill¹

¹ Fred Hutchinson Cancer Center, Seattle, United States

Abstract Background: Association of bacterial taxa to graft vs. host disease (GVHD) have relied on examination of fecal samples, but tissue and cellular localization remains largely unexplored due to technical challenges. We augment a spatial transcriptomics (ST) platform to address this limitation. Identification of specific bacterial taxa in the context of ST has primarily been employed using sequencing-based platforms due to the need to provide taxonomic resolution of the 16S ribosomal RNA that traditionally requires at least 200 base pairs for sequence discrimination. However, commercially available platforms that utilize this approach (e.g. 10x Visium) do not have the capability for achieving sub-cellular imaging resolution. Imaging-based platforms (e.g. 10x Xenium) can localize short (~40 base pair) oligonucleotide probes with sub-cellular resolution but do not always discriminate between relevant bacterial species. Here we devise a computational approach to generate short oligonucleotide panels that can identify and localize gastrointestinal (GI)-specific microbial taxa with sub-cellular resolution, which compatible with standard ST human gene panels, and can be run on formalin-fixed paraffin embedded tissue samples (FFPE).

Methods: We selected a panel of 20 bacterial species that are commonly found to dominate the post-transplant GI microbiota or to have been correlated with transplant outcomes. To identify 40 base pair 16S rRNA oligonucleotides specific for our species of interest, we ran a BLAST search consisting of all possible 40 base pair sliding windows against a database of 2,685 whole bacterial genomes for off-target prediction. Bacterial genomes were selected for off-target screening if found in greater than 0.01% abundance using the publicly available GMrepo v2 gut microbiota database. At least 2 probes with high-specificity and no off-target overlap for each bacterial species were selected for inclusion in custom panel design. We validate our custom probe set in addition to the 322 human colon gene set using the 10x Xenium platform on 10 endoscopic biopsy FFPE tissue sections derived from 5 post-allogeneic stem cell transplant patients with or without graft-vs.-host disease (GVHD) (severe N=3, none N=2). To improve exposure of intracellular bacterial contents, each patient sample was run in duplicate to test the addition of lysozyme to the standard ST workflow. As control, 16S rRNA gene sequencing was performed on stool brushings for all patients.

Results: We detected transcript dominance of four taxa commonly associated with GVHD, including *E. faecalis* and *E. coli* in patients with severe GVHD as well as *A. muciniphila* and *B. fragilis* in patients without GVHD. This was confirmed with 16S rRNA gene stool brushing sequencing, where the above 4 organisms were found at 17%, 13%, 26%, and 14% relative abundance, respectively, in distinct patient samples. In colonic tissue with severe GVHD, domination with *E. faecalis* and *E. coli* were detected and bacterial transcripts predominantly co-localized with colonic enterocytes (*E. faecalis* 58%, *E. coli* 46%), followed by activated fibroblasts (*E. faecalis* 23%), and enteroendocrine cells (*E. coli* 12%). In colonic tissue with no GVHD, domination with *A. muciniphila* was detected and transcripts primarily co-localized with pericryptal fibroblasts (56%), macrophages (12%), and goblet cells (10%). Similarly, domination with *B. fragilis* demonstrated transcripts that colocalized with macrophages (38%) and fibroblasts (19%). Addition of lysozyme improved bacterial detection in FFPE colon tissue samples for both gram positive and negative bacterial species, with an average increase of 1,161 to 2,643 transcripts per tissue section.

Conclusions: We demonstrate the feasibility of using short 40 base pair oligonucleotide probes to detect targeted bacterial species in the context of the GI microbiota on a gene multiplex ST platform. The ability to pair custom bacterial probes on top of existing human ST gene panels enables examination of the GI microbiota-immune interface on a subcellular level. Here, we show that bacterial taxa associated with GVHD are spatially resolvable and colocalize to distinct cell types.

<https://doi.org/10.1182/blood-2025-2311>